

Timeline Check In

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Setup

Packages

Data

Ideas for analysis and background

Our project focuses on salt marsh vegetation at North Campus Open Space (NCOS), with the goal of understanding which plant species characterize this restored wetland habitat and how those patterns have changed across survey years. Rather than treating vegetation cover as only a set of numbers, we are using percent cover to describe how the salt marsh is developing as a plant community over time. In this sense, our project asks how restoration becomes visible through the species that take root, persist, expand, or decline across the marsh.

Salt marsh vegetation is especially useful for understanding restoration trajectory because these plant communities are shaped by strong environmental filters, including salinity, tidal flooding, saturated soils, and seasonal changes in water availability. In southern California salt marshes, native halophytes such as pickleweed (*Salicornia pacifica*) and saltgrass (*Distichlis spicata*) often help define the structure and identity of the habitat (Ferren 1985; North Campus Open Space 2025). Because these species are adapted to stressful coastal wetland conditions, changes in their mean percent cover can help indicate whether the restored salt marsh is developing toward a recognizable native marsh community.

This question is particularly relevant at NCOS because the site was restored from the former Ocean Meadows Golf Course into a mosaic of wetland and upland habitats, including salt marsh, brackish marsh, seasonal wetlands, and transitional habitats (Cheadle Center for Biodiversity and Ecological Restoration and Environmental Science Associates 2016; Rickard 2024). Broader NCOS monitoring reports evaluate restoration progress across habitat types, while our project narrows in on salt marsh vegetation specifically. By focusing on dominant

species and native/non-native cover through time, we can connect the survey data to larger questions about restoration progress, habitat identity, and community development.

Previous work on California wetland restoration has shown that vegetation establishment can be slow, variable, and strongly shaped by local site conditions such as elevation, soil salinity, soil moisture, planting, and adaptive management (Beheshti et al. 2023). This matters for our analysis because shifts in dominant plant cover are not just visual patterns in a graph. They may reflect ecological processes such as native plant establishment, non-native plant expansion, hydrologic change, disturbance, or interannual variability in rainfall and growing conditions.

Ecological background

Salt marshes are dynamic coastal wetlands where plant communities exist at the edge of land and water. Their vegetation patterns are shaped by both physical stress and biological interactions, making them useful indicators of environmental conditions and restoration progress. In southern California, seasonal rainfall and dry summers can influence freshwater input, soil moisture, salinity, and plant growth, which may affect how vegetation cover changes from year to year (Ferren 1985).

At NCOS, we expect dominant salt marsh species to include native salt-tolerant plants such as *Salicornia pacifica* and *Distichlis spicata*. The NCOS coastal salt marsh habitat page describes pickleweed as usually the dominant species in coastal salt marshes and also identifies saltgrass, alkali heath, fleshy jaumea, and California saltbush as characteristic salt marsh plants (North Campus Open Space 2025). These expectations give us a useful ecological reference point for interpreting our data: if the restored salt marsh is developing toward a recognizable coastal salt marsh community, we would expect these native halophytes to contribute strongly to vegetation cover.

Native and non-native cover are also important to examine because restoration success depends not only on whether plants are present, but on which plants are shaping the habitat. A marsh with high total vegetation cover may still be ecologically different depending on whether that cover is dominated by native salt marsh species or by non-native plants. NCOS monitoring reports describe vegetation cover, native biodiversity, and relative native cover as important indicators of restoration progress (Rickard 2024). Our project builds from that monitoring context by focusing more closely on the species-level patterns that contribute to salt marsh identity through time.

Planned analyses

Data cleaning and wrangling

Summaries for planned analyses

Visualizations directly relevant to answering questions

Figure 1: Dominant species percent cover across survey years

```
fig1 <- ggplot(data = dominant_by_year,
               aes(x = year,
                   y = mean_pc,
                   fill = psoc_grouped)) +
  geom_col(position = "stack") +
  scale_fill_manual(values = c(
    "Salicornia_pacifica"      = "#0D5E45",
    "Distichlis_spicata"      = "#1D9E75",
    "Atriplex_lentiformis"    = "#5DCAA5",
    "Eleocharis_macrostachya" = "#A8DBC9",
    "Raphanus_sativus"       = "#D85A30"
  )) +
  scale_x_continuous(breaks = seq(2018, 2025, 1)) +
  labs(x = "Survey year",
       y = "Mean percent cover (%)",
       fill = "Species") +
  theme_minimal() +
  theme(legend.position = "bottom",
        legend.text = element_text(face = "italic"),
        legend.title = element_text(face = "bold"))

fig1
```

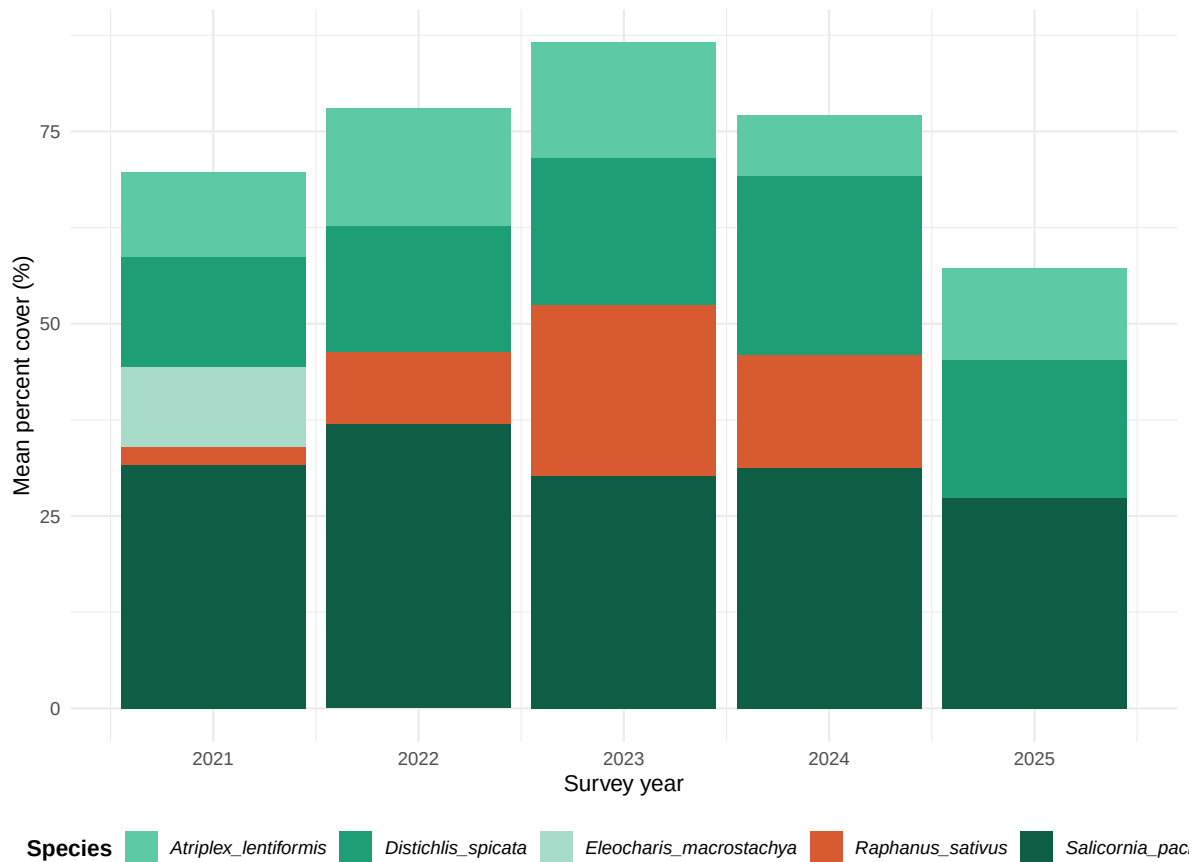


Figure 1: Mean percent cover of the five dominant salt marsh plant species at NCOS across survey years, averaged across transects within each year. Dominant species were identified as the top five by mean percent cover across all survey years. Bar height reflects total mean cover of the five dominant species and varies across years. Green shades indicate native species; coral indicates non-native species.

Figure 2: Proportional native vs. non-native cover across survey years

```
fig2 <- ggplot(data = native_nonnative,
               aes(x = year,
                   y = prop,
                   fill = cover_category)) +
  geom_col(position = "stack") +
  scale_fill_manual(values = c("NATIVE COVER" = "#1D9E75",
                              "NON-NATIVE COVER" = "#D85A30")) +
  scale_y_continuous(labels = scales::percent) +
  labs(x = "Survey year",
```

```

y = "Proportion of vegetation cover",
fill = "Cover category") +
theme_minimal() +
theme(legend.position = "bottom")

```

fig2

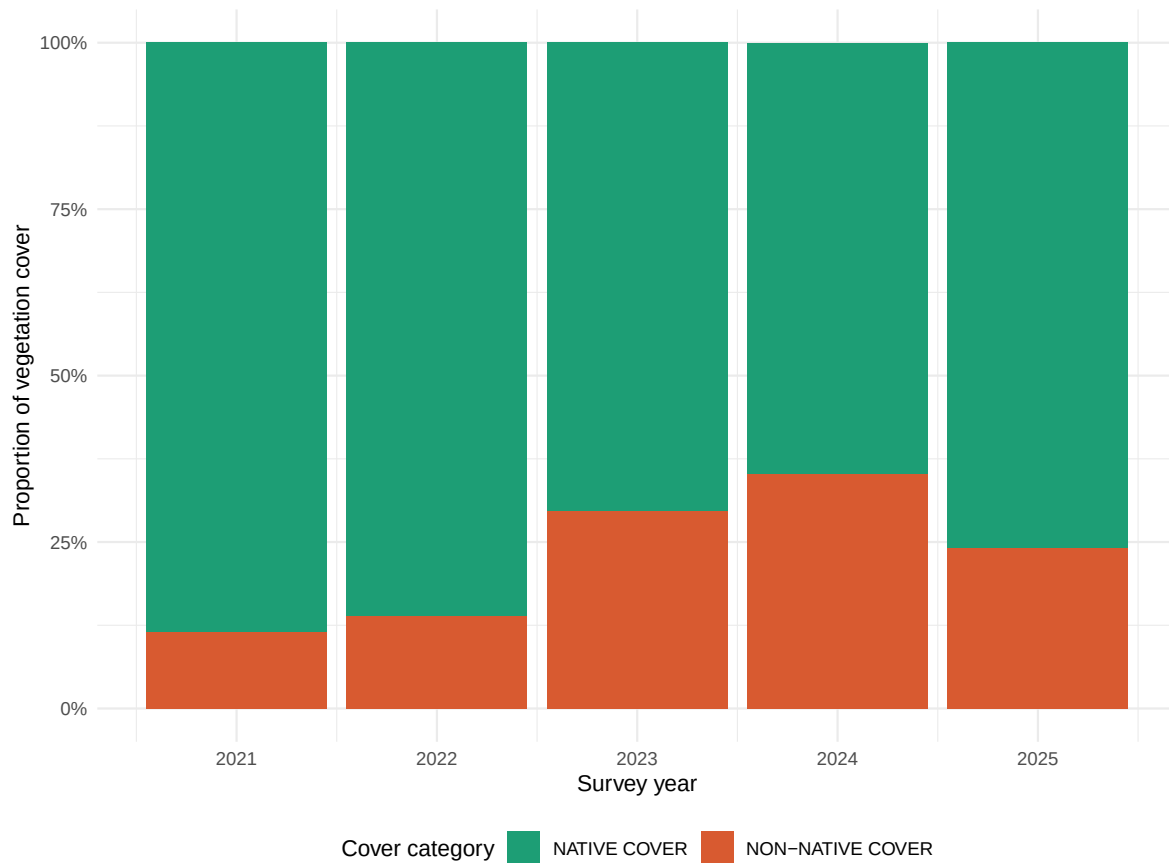


Figure 2: Proportional contribution of native and non-native vegetation cover in salt marsh habitat at NCOS across survey years. Each bar sums to 1. Native cover is shown in green; non-native cover in coral.

Figure 3: Dominant non-native species percent cover across survey years

```

#Creating a red/coral color palette for the top non-native species
nonnative_palette <- c("#7F1D1D", "#B8322A", "#D85A30", "#E9825A", "#F3B49C")

```

```

#Visualizing top non-native species over time
fig_3 <- ggplot(data = top_nonnative_by_year,
               aes(x = year,
                   y = mean_pc,
                   fill = psoc)) +
  geom_col(position = "stack") +
  scale_fill_manual(values = nonnative_palette) +
  scale_x_continuous(breaks = seq(2018, 2025, 1)) +
  labs(x = "Survey year",
       y = "Mean percent cover (%)",
       fill = "Non-native species") +
  theme_minimal() +
  theme(legend.position = "bottom",
        legend.text = element_text(face = "italic"),
        legend.title = element_text(face = "bold"))

```

fig_3

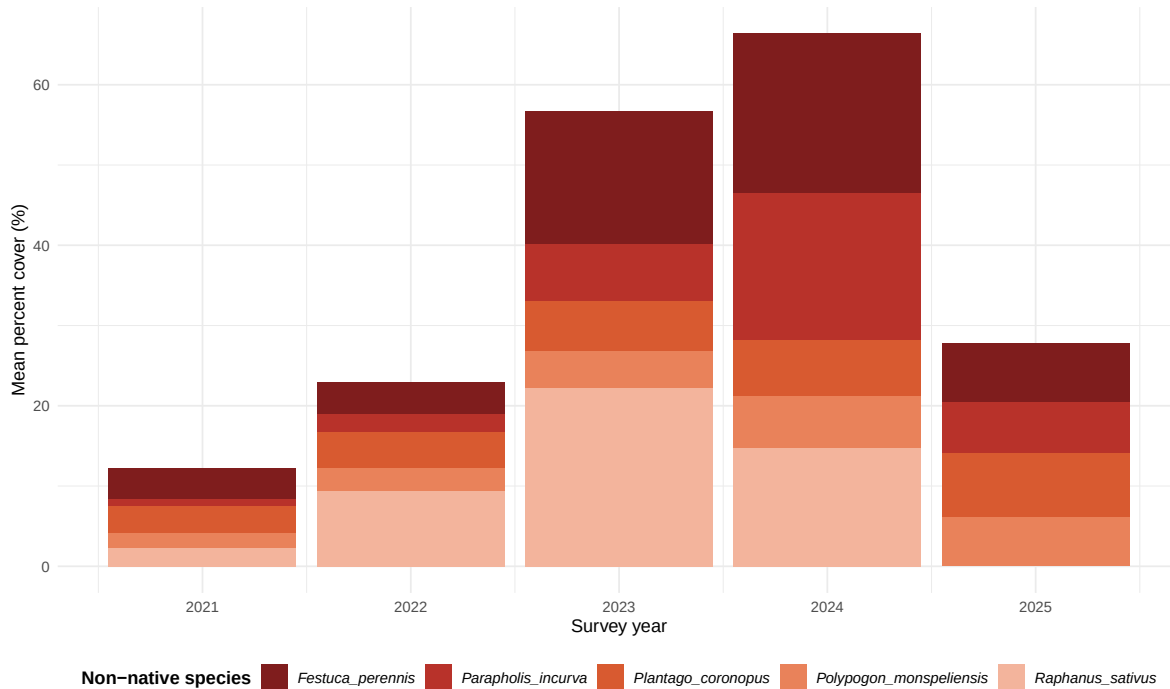


Figure 3: Mean percent cover of the five most common non-native plant species in the filtered salt marsh/brackish marsh dataset across survey years, averaged across transects within each year. Bar height reflects the summed mean cover of these non-native species in each year.

Figure 4: Species rank plot

```
# fig3 - pointrange instead of col
cutoff_x <- mean(c(species_rank$mean_pc[5], species_rank$mean_pc[6]))

fig4 <- ggplot(data = species_rank_top20,
               aes(x = mean_pc,
                   y = reorder(psoc, mean_pc),
                   color = cover_category)) +
  geom_pointrange(aes(xmin = mean_pc - se_pc,
                      xmax = mean_pc + se_pc)) +
  geom_vline(xintercept = cutoff_x,
             linetype = "dashed",
             color = "gray50") +
  scale_color_manual(values = c("NATIVE COVER" = "#1D9E75",
                                "NON-NATIVE COVER" = "#D85A30")) +
  labs(x = "Mean percent cover ( $\pm$  1 SE)",
```

```

y = "Species",
color = "Cover category") +
theme_minimal() +
theme(axis.text.y = element_text(face = "italic", size = 9),
      legend.position = "bottom",
      legend.title = element_text(face = "bold"))

```

fig4

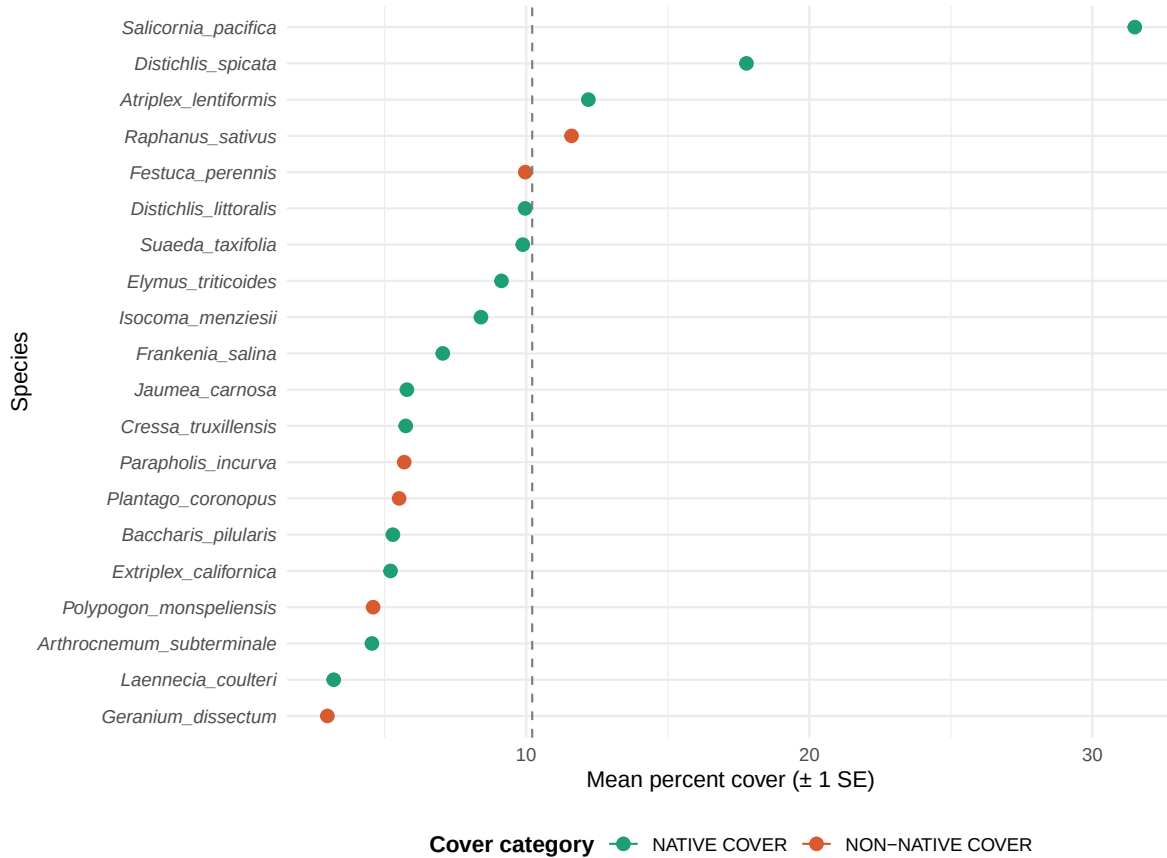


Figure 4: Mean percent cover (± 1 SE) of the top 20 plant species recorded in salt marsh habitat at NCOS, averaged across transects and survey years and ranked from highest to lowest mean cover. Points represent means and horizontal bars represent ± 1 standard error. Green indicates native species; coral indicates non-native species. Dashed line marks the cutoff for the five dominant species visualized in figure 1.

Other exploratory visualizations

Figure 5: Total vegetation cover across survey years

```
fig5 <- ggplot(data = total_cover,
               aes(x = year,
                   y = mean_total)) +
  geom_ribbon(aes(ymin = mean_total - se_total,
                 ymax = mean_total + se_total),
             fill = "#1D9E75",
             alpha = 0.2) +
  geom_line(color = "#1D9E75",
            linewidth = 1) +
  geom_point(color = "#1D9E75",
             size = 3) +
  labs(x = "Survey year",
       y = "Total mean percent cover (%)") +
  theme_minimal()

fig5
```



Figure 5: Total mean percent vegetation cover (native + non-native combined) in salt marsh habitat at NCOS across survey years. Shaded band represents ± 1 standard error around the mean.

Plan for elective (if completing)

Elective concept

Draft sections (include some description/image of drafts)

Plan for the next two weeks

Next steps

References

Beheshti, Kathryn M., Stephen C. Schroeter, Andres A. Deza, Daniel C. Reed, Rachel S. Smith, and Henry M. Page. 2023. "Large-Scale Field Studies Inform Adaptive Management of California Wetland Restoration." *Restoration Ecology* 31 (5): e13936. <https://doi.org/10.1111/rec.13936>.

Cheadle Center for Biodiversity and Ecological Restoration, and Environmental Science Associates. 2016. *North Campus Open Space Restoration Project Restoration Plan*. University of California, Santa Barbara. <https://escholarship.org/uc/item/9505h60h>.

Ferren, Wayne R. Jr. 1985. *Carpinteria Salt Marsh: Environment, History, and Botanical Resources of a Southern California Estuary*. Publication Number 4. The Herbarium, Department of Biological Sciences, University of California, Santa Barbara.

North Campus Open Space. 2025. *Coastal Salt Marsh*. <https://www.ncos.ccber.ucsb.edu/native-plant-habitats/coastal-salt-marsh>.

Rickard, Alison. 2024. *North Campus Open Space Restoration Project Monitoring Report: Year 7, December 2024*. University of California, Santa Barbara Cheadle Center for Biodiversity; Ecological Restoration. <https://escholarship.org/uc/item/7bq618m8>.